Test Report

Applicant:	Xuzhou Huizhixin Glass Products Co., Ltd.
Address:	No. A1-55, Mapo Town Industrial Park, Tongshan District, Xuzhou City, Jiangsu Province
Manufacturer:	Xuzhou Huizhixin Glass Products Co., Ltd.
Address:	No. A1-55, Mapo Town Industrial Park, Tongshan District, Xuzhou City, Jiangsu Province

Report on the submitted samples said to be:

Product Name:	tumbler
Brand Name:	N/A
Model Number:	Vertical striped glass 600ML Vertical striped glass 400ML, Stemmed glass, Cup with a round - bead handle, Flat - bottomed breakfast cup, Breakfast cup with a handle
Date of Receipt:	Nov. 26, 2024
Date of Test:	Nov. 26, 2024 ~ Dec. 02, 2024
Date of Report:	Dec. 05, 2024
Test Method:	Please refer to next page.
Test Result:	Please refer to next page.

Prepared (Engineer): Slien Wang

Reviewer (Supervisor): Xiaoshan Ni

This test report is based on a single evaluation of one sample of above mentioned products. It is not permitted to be duplicated in extracts without written approval of Shenzhen DL Testing Technology Co., Ltd.

Version

Version No.	Date	Description
00	Dec. 02, 2024	Original
01	Dec. 05, 2024	This report modified the description of the sample model on the basis of the original report(DL-241126088RR), and the original report has been invalid, please note.

Test Requested: Conclusion

German Food, Articles of Daily Use and Feed Code of September 1, 2005 (LFGB) Section 30 and 31 With amendments,Regulation 1935/2004/EC on materials and articles intended to come into contact with food,German, European Commission Regulation (EU) No. 10/2011 with amendments and BfR recommendation, European Commission Regulation AP(2004)5,European Commission Regulation 84/500/EEC and its Amendment 2005/31/EC,French DGCCRF 2004-64 and French Décret n°92-631, European Commission Regulation CM/Res(2020)9.

1.Sensory test-taste and odour to the integrate product	Pass
2.Visible Color Migration	Pass
3.Overall Migration Test	Pass
4.Polycyclic Aromatic Hydrocarbons (PAHs)	Pass
5.Volatile Organic Matter（VOM）	Pass
6.Specific Migration of Polycyclic Aromatic Hydrocarbons (PAHs)	Pass
7.Total Lead (Pb)、Cadmium (Cd)、Zinc (Zn)、Platinum (Pt)	Pass
8.Peroxides Value	Pass
9.Extractive Substance	Pass
10.Organic Tin content	Pass
11.Lip and rim region: Leachable Lead and Cadmium	Pass
12.Lip and rim region: Leachable Cobalt	Pass
13.Leachable Lead、Cadmium、Cobalt、Aluminum and Arsenic	Pass

Test Part Description:

Specimen No.	Description	Material
01	Grey silicone (pipette)	silicone
02	Transparent glass (cover)	glass
03	Transparent glass (cup)	glass
04	Transparent glass (straw)	glass

Test Results:

1.Sensory test-taste and odour to the integrate product

Test Method: With reference to Robinson's test with reference to DIN 10955:2004.
Test conditions: Distilled wate, 100℃, 2h

Test Item	01	02	03	04	Limit
Sensorial examination odour (Point scale)	1	0	0	0	3
Sensorial examination taste (Point scale)	1	0	0	0	3

Scale evaluation:

0: No perceptible odour and taste

1: Odour and taste just perceptible (still difficult to define)

2: Moderate odour and taste

3: Moderate strong odour and taste

4: Strong odour and taste

2.Visible Color Migration

Test Method :With reference to LFGB 30&31, BfR IX Visible Color Migration.

Test Items	Test condition	01	Limit
Visible Color Migration	3% Acetic Acid Aqueous solution (70℃, 2hours)	Absent	Absent
Visible Color Migration	10% Ethanol Aqueous solution (70℃, 2hours)	Absent	Absent

3.Overall Migration Test

Test Method: With reference to EN 1186-1: 2002 and EN 1186-3: 2022.

Simulant Used	Time	Temperature	Unit	Limit	01		
					1st	2nd	3rd
3% Acetic acid	2.0h	70℃	mg/dm^2	10	N.D.	N.D.	N.D.
10% Ethanol	2.0h	70℃	mg/dm^2	10	N.D.	N.D.	N.D.
95% ethanol	2.0h	60℃	mg/dm^2	10	N.D.	N.D.	N.D.
Isooctane	0.5h	40℃	mg/dm^2	10	N.D.	N.D.	N.D.

Note:

(1) mg/dm^2 = milligram per square decimeter;

(2) < =less than;

(3) N.D. = Not Detected (<Limit).

4.Polycyclic Aromatic Hydrocarbons (PAHs)

Test Method: With reference to AfPS GS 2019:01 PAK, by solvent extraction and analysis was performed by gas chromatographic-mass spectrometer (GC-MS).

Item	Unit	MDL	01			Limit
			1st	2nd	3rd	
Naphthalene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Phentherene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Anthracene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Fluoranthen	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[a]anthracene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Chrysene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[b]fluoranthene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[k]fluoranthene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[a]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Indeno[1,2,3-cd]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Dibenzo[a,h]anthracene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[g,h,i]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[j]fluoranthene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[e]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Total of 15 PAHs	mg/kg	/	N.D.	N.D.	N.D.	/

Note:
(1) 1mg/kg=0.0001%;
(2) < =less than;
(3) N.D. = Not Detected (<MDL);
(4) MDL= Method Detection Limit.

5.VOM-BfR Besummung von fluchtigen Verbindungen in Bedarfsgegenstanden aus silicone Version2 Stand: 09/2023

Test Method: With reference to 19. Mitteilung über die Untersuchung von Kunststoffen. Bundesgesundheitsblatt 14(1971)265

Test condition	MDL (%)	Limit (%)	01
200℃，4h	0.1	0.5	0.2

Note:
(1) 1mg/kg=0.0001%;
(2) < =less than;
(3) N.D. = Not Detected (<MDL);
(4) MDL= Method Detection Limit.

6.Specific Migration of Polycyclic Aromatic Hydrocarbons (PAHs)

Test Method: With reference to BS EN 13130-1: 2004, determined by GC-MS.
Test conditions: 3% acetic acid, 70℃, 2h

Item	Unit	MDL	01			Limit
			1st	2nd	3rd	
Naphthalene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Phentherene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Anthracene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Fluoranthen	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[a]anthracene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Chrysene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[b]fluoranthene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[k]fluoranthene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[a]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Indeno[1,2,3-cd]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Dibenzo[a,h]anthracene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[g,h,i]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[j]fluoranthene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Benzo[e]pyrene	mg/kg	0.1	N.D.	N.D.	N.D.	Absent
Total of 15 PAHs	mg/kg	/	N.D.	N.D.	N.D.	/

Note:
(1)1mg/kg=0.0001%;
(2)N.D. = Not Detected (<MDL);
(3)MDL= Method Detection Limit.

7.Total Lead (Pb)、Cadmium (Cd)、Zinc (Zn)、Platinum (Pt)

Test Method:With reference to EPA METHOD 3052:1996, analyzed by ICP-OES.

Test Item	Unit	MDL	Limit	01
Lead (Pb)	mg/kg	5	100	N.D.
Cadmium (Cd)	mg/kg	5	100	N.D.
Zinc (Zn)	mg/kg	5	100	N.D.
Platinum (Pt)	mg/kg	5	50	N.D.

Note:
(1)1mg/kg=0.0001%;
(2)N.D. = Not Detected (<MDL);

(3)MDL= Method Detection Limit.

8.Peroxides Value

Test Method:With reference to European Pharmacopeia, Ph.Eur.Method 2.5.5

Test Item	Limit	01			Conclusion
		1st	2nd	3rd	
Peroxides value	Absent	Absent	Absent	Absent	PASS

9.Extractive Substance

Test Method:The test was performed according to the 61st Communication on testing of plastics Bundesgesundheitsbl.46(2003)362

Simulant Used	Test duration /Temperature	Unit	Limit	01		
				1st	2nd	3rd
Distilled water	Reflux for 5 hours	%	0.5	<0.1	<0.1	<0.1
3% Acetic acid	Reflux for 5 hours	%	0.5	<0.1	<0.1	<0.1
10% Ethanol	Reflux for 5 hours	%	0.5	<0.1	<0.1	<0.1

Note:

(1) 1mg/kg=0.0001%;

(2) < =less than.

10.Organic Tin content

Test method :With reference to ISO 17353 :2004, analysis was performed by GC-MS.

Test Item	Unit	MDL	Limit	01
Dibutyltin (DBT)	mg/kg	0.01	1	N.D.
Tributyltin (TBT)	mg/kg	0.01	1	N.D.
Triphenyltin (TPT)	mg/kg	0.01	1	N.D.
Dioctyltin（DOT）	mg/kg	0.01	1	N.D.
Monobutyltin (MBT)	mg/kg	0.01	1	N.D.
Monooctyltin (MOT)	mg/kg	0.01	1	N.D.
Tetrabutyltin (TTBT)	mg/kg	0.01	1	N.D.

Note:

(1) 1mg/kg=0.0001%;

(2) N.D. = Not Detected (<MDL);

(3) MDL= Method Detection Limit.

11.Lip and rim region: Leachable Lead and Cadmium

Test Method:With reference to DIN 51032-2017 and EN 1388-2:1995,determined by ICP-OES.

Test conditions: 4% Acetic acid, 70℃, 24h

Test Items	MDL	Limit	03
Lead(Pb)	0.1mg/dm^2	0.4mg/dm^2	N.D.
Cadmium(Cd)	0.01mg/dm^2	0.05mg/dm^2	N.D.

Note:

(1) mg/dm^2 = milligram per square decimeter;

(2) N.D. = Not Detected (<MDL).

Flatware: With an internal depth not exceeding 25mm, measured from the lowest point to the horizontal plane passing through the overflow point.

Hollowware: Those articles which do not fall into the category of Flatware.

Drinking rim: The drinking rim is a 20mm wide section of the external surface of a drinking vessel. The width is measured downwards from the upper edge along the wall of the vessel.

12.Lip and rim region: Leachable Cobalt

Test Method:With reference to ALS Opinion Number 2017/15 and EN 1388-2:1995,determined by ICP-OES.

Test conditions: 4% Acetic acid, 70℃, 24h

Test Items	MDL	Limit	03
Cobalt(Co)	0.01mg/dm^2	0.02mg/dm^2	N.D.

Note:

(1) mg/dm^2 = milligram per square decimeter;

(2) N.D. = Not Detected (<MDL).

13.Leachable Lead 、Cadmium and Cobalt

Test Method :With reference to ISO 6486-1:2019,determined by ICP-OES.

Test conditions: 4% Acetic acid, 70℃, 24h

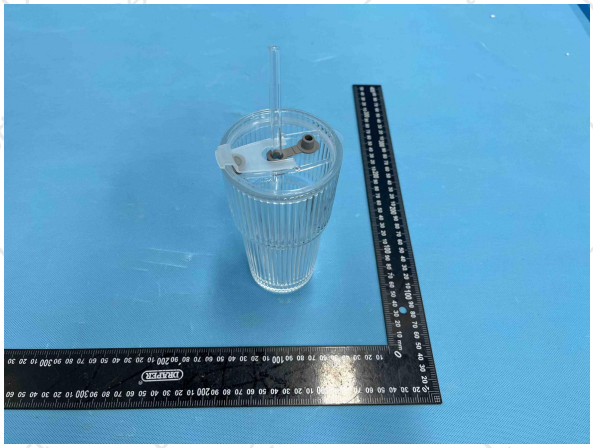
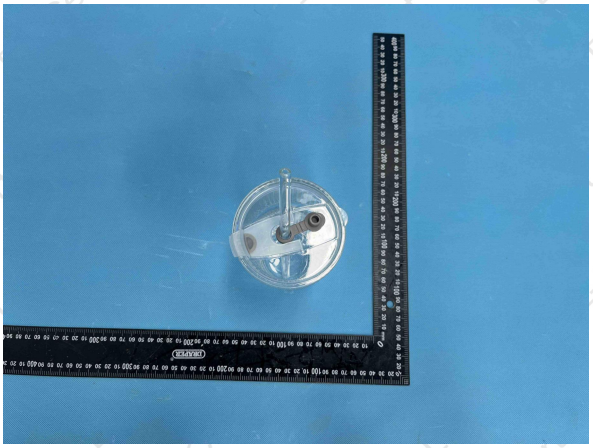
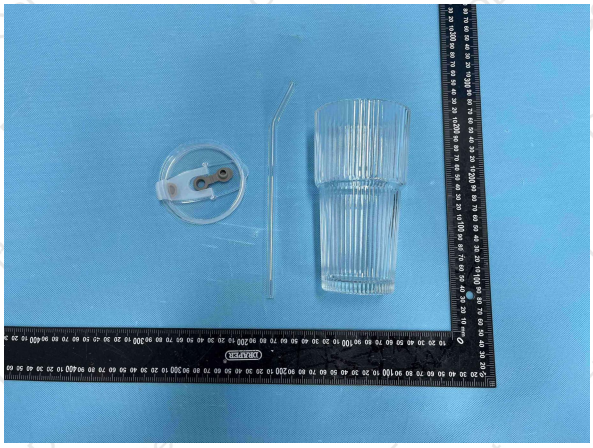
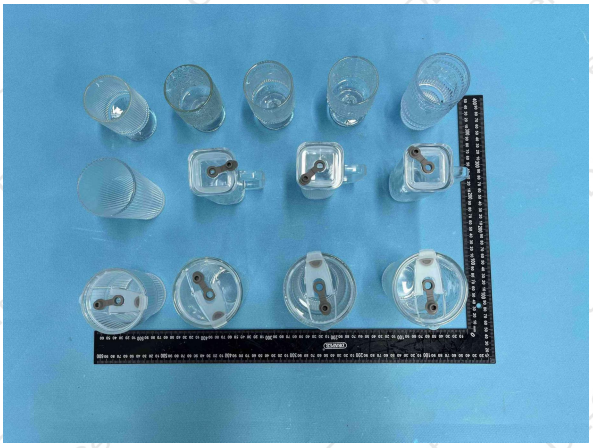
Test Items	MDL	02	03	04
Lead(Pb)	0.1mg/L	N.D.	N.D.	N.D.
Cadmium(Cd)	0.01mg/L	N.D.	N.D.	N.D.
Cobalt(Co)	0.1mg/L	N.D.	N.D.	N.D.
Aluminum(Al)	0.1mg/L	N.D.	N.D.	N.D.
Arsenic(As)	0.1mg/L	N.D.	N.D.	N.D.

Note:

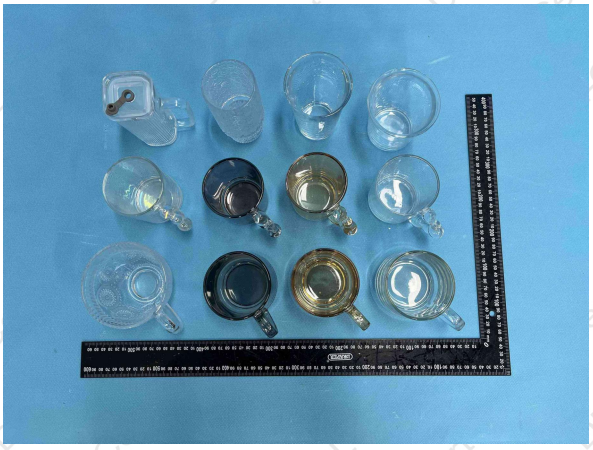
(1) mg/L= milligram per liter;

(2) N.D. = Not Detected (<MDL).

Photograph of Sample





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